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Computational modeling of laser welding for aluminum–copper joints using a circular strategy

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ABSTRACT

Dissimilar metal welding poses a significant challenge for researchers in the welding and materials science communities due to the different properties of the metals involved. In this study, we present a novel approach for determining the optimal processing parameters for continuous laser welding (CLW) of dissimilar metals, specifically aluminum and copper, using a circular strategy. The approach utilizes computational fluid dynamics (CFD) simulations to model the welding process and predict joint quality. We validated a CFD-based numerical model for predicting the melt-pool shape across a range of laser power and circular welding speed values. Additionally, we developed artificial neural network (ANN) models that can predict melt pool interface width, penetration depth, and copper concentration for any combination of processing conditions. Using surrogate modeling, we identified the optimal values of laser power and welding speed that result in the highest quality joint. Our results showed that the optimal processing conditions were a laser power of 2950W and a circular speed of 55 mm/s, which resulted in a high-quality Al–Cu joint with maximum shear strength of 2200 N. Microstructural analysis and tensile-shear strength testing confirmed that the optimal conditions resulted in a low degree of intermixing and the absence of large intermetallic compounds (IMCs) in the weld seam. Overall, study offers a valuable contribution to the welding and materials science communities by presenting a novel approach to address a common challenge in dissimilar metal welding. Our approach can be applied to other types of dissimilar metal welding and may inform the development of new welding techniques in the future.

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1. Introduction

The optimization of welding dissimilar materials has been a crucial aspect of modern industrial production in recent decades. The demand for products with improved characteristics such as lighter weight, increased stiffness, strength and resistance to corrosion and high temperatures has led to the integration of hybrid, composite, and dissimilar materials in manufacturing processes. Joining different metallic materials using both traditional and new welding techniques is a key aspect in the development of innovative manufacturing solutions. The implementation of dissimilar metal combinations can offer numerous advantages in enhancing the electrical, mechanical, thermal, and corrosion resistance properties of connections [1]. However, conventional welding methods, such as ultrasonic welding [2] and resistance spot welding [3], have limitations including lack of control over heat input,

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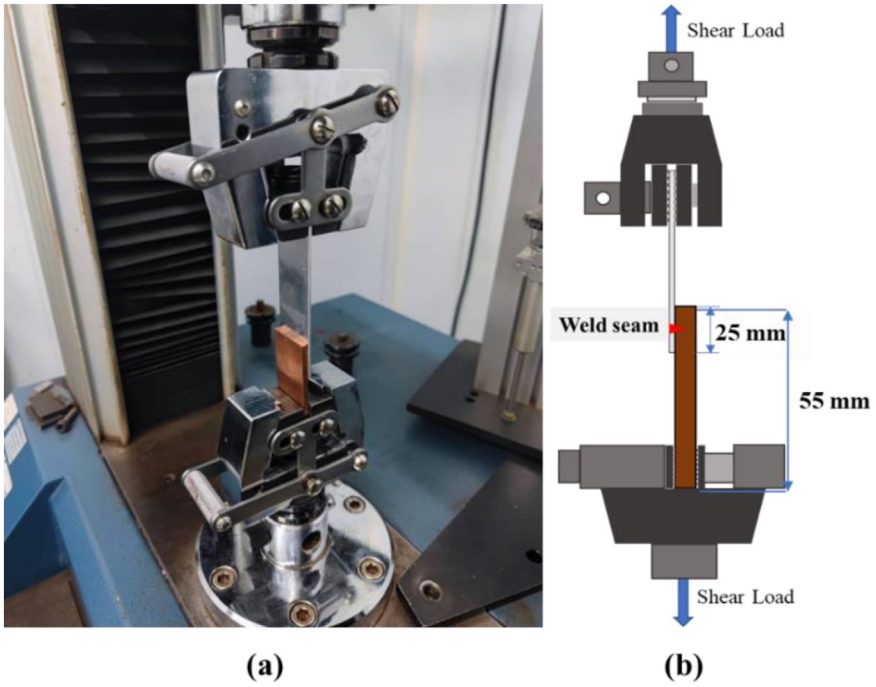


Fig. 3 – Tensile test for Al–Cu samples (a) Setup for experiment and (b) Specimen dimension.

experimental investigations. This adjustment was implemented to enhance the reliability and accuracy of the simulation results.

3.1. *Governing equations*

The fluid flow is the determining factor for the mixing of metals within the melt pool [19]. In the present simulations, the equations used for calculating the geometry of the melt pool and the concentration of Cu in both the solid and liquid phases are as follows [17].

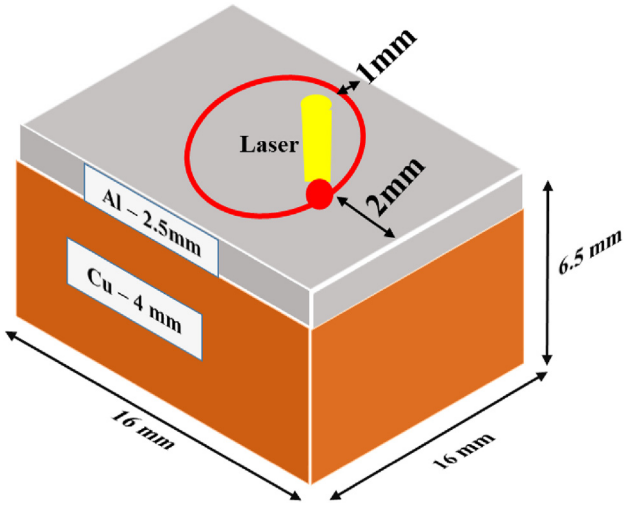


Fig. 4 – Geometry for simulation domain.

- Mass conservation equation

$$\frac{\partial}{\partial t}(\rho)+\nabla\left(\rho\overrightarrow{V}\right)=0 \tag{1}$$

- Momentum conservation equation

$$\frac{\partial}{\partial t}\left(\rho\overrightarrow{V}\right)+\nabla\left(\rho\overrightarrow{V}\overrightarrow{V}\right)=\nabla\left(\mu\nabla\overrightarrow{V}\right)-\nabla P-\frac{\mu}{K}\overrightarrow{V}+\rho g \tag{2}$$

- Energy conservation equation

$$\frac{\partial}{\partial t}(\rho h)+\nabla\left(\rho\overrightarrow{V}h\right)=\nabla(k\nabla T) \tag{3}$$

- Species conservation equation

$$\frac{\partial}{\partial t}(\rho Y)+\nabla\left(\rho\overrightarrow{V}Y\right)=\nabla(D\rho\nabla Y) \tag{4}$$

Equation (1) represents the relationship between density of the fluid denoted by ρ, and the fluid velocity $\overrightarrow{V}$, and Eq. (2), on the other hand, relates the dynamic viscosity of fluid μ to the pressure P. Additionally, the isotropic permeability is represented by K (as expressed by the Kozeny-Carman equation [20]), and the acceleration due to the gravity is g. It is noted that the other detail in derivations can be found in Ref. [13].

The VOF method was utilized to track the evolution of the keyhole interface, including the melt pool surface and

keyhole wall. The governing equation for this method was expressed as:

$$\frac{\partial f}{\partial t} = \nabla\left(f.\vec{V}\right) = 0 \tag{5}$$

where f represents the fraction of the liquid volume over the entire cell volume, and $\vec{V}$ denotes the fluid velocity obtained from the solutions of Eqs. (1) and (2). It is important to note that the volume of fluid (VOF) method involves the use of a parameter denoted as f which takes on a value of 1 for cells that are in either a liquid or solid state, 0 for cells that are in a gaseous state, and a value between 0 and 1 for cells that are in an intermediate state. The value of f indicates the volume fraction of liquid present in a particular cell, with higher values indicating a greater volume fraction of liquid [17]. Throughout the simulation process, Eq. (5) is updated at each time step, and the keyhole interface is identified as the set of cells for which $1 > f > 0$ [17]. Thus, these cells represent the region of the material undergoing melting or vaporization. The formulation of the VOF method is a well-established technique in computational fluid dynamics for modeling multiphase flows, and has been applied to a variety of engineering problems, including laser welding and cutting.

3.2. Driving forces

The primary forces that govern the behavior of the molten pool during welding are gravity, recoil force, buoyancy force, surface tension force, and shear stress generated by the flow of metallic vapor. The keyhole phenomenon, which occurs when the laser power is high, is a critical aspect of deep penetration welding [21]. The target material undergoes melting and subsequent evaporation when subjected to a high-intensity laser beam, which generates pressure and results in the formation of a keyhole, a hollow cavity filled with vapor, through liquid ejection. The recoil pressure has been computed in the present simulations as follows [17].

$$P_r = \alpha P_v \exp\left[\frac{\Delta H_v}{(\gamma_v - 1)C_v T_v}\left(1 - \frac{T_v}{T}\right)\right] \tag{6}$$

where α is the evaporation coefficient, P_v is the evaporation pressure, the latent heat of evaporation ΔH_v, the heat capacity ratio of the metal vapor γ_v, the specific heat of the metal vapor C_v, the evaporation temperature T_v, and the temperature at the keyhole wall T. Specifically, the evaporation recoil pressure is directly proportional to the saturated vapor pressure and is influenced by the specific properties of the materials being used. The further detail in this section about driving forces and be found in Refs. [13,17].

3.3. Volume fraction of different materials

In the simulations conducted, the fluid flow was comprised of a mixture of two distinct metals - aluminum (Al) and copper (Cu). In order to assess the behavior of the melt pool that was formed as a result of this flow, we utilized a mixed volume fraction model to calculate the average physical properties of the flow. The model is described as follows:

$$\overline{\Phi} = (1 - f_2)\Phi_2 + f_2\Phi_1 \tag{7}$$

where $\overline{\Phi}$ is the average value of the physical property of interest, Φ_1 is the corresponding physical property of Material #1, Φ_2 is the physical property of Material #2, and f_2 is the relative volume fraction of the two materials.

3.4. Boundary conditions

The boundary conditions for the simulation were defines as [22], the convection, radiation and evaporation was considered for the top free surface. Moreover, the pressure boundary condition for the top free surface were modeled using Young-Laplace equation, which relates the pressure difference across an interface between two fluids to the curvature of the interface. However, for other surfaces the convective heat transfer and radiative heat transfer were considered to model the temperature distribution and heat transfer at a surface in contact with a fluid or gas. All the details about the boundary conditions on different surface can be found in Refs. [13,17].

3.5. Material properties

In this study, the temperature-dependent physical properties of Al and Cu metals such as thermal conductivity, specific heat, and density were considered and accounted for in the simulation. Unlike previous studies that assumed constant physical properties, the current study used the physical properties with respect to the temperature T(K). The detail of those material properties can be found in Refs. [13,17].

3.6. Simulation validation

To validate the accuracy of the computational model, a comparative analysis was conducted between the model's predictions and experimental outcomes for the interface width and penetration depth of the melt pool in single scan-track experiments conducted under different laser powers and circular speeds. The comparison between the experimental and simulation results for the boundary regions of the melt pool, depicted in Fig. 5, demonstrated that the simulation outcomes exhibit good qualitative agreement with the experimental data. Furthermore, no evidence of cracks or defects in the melt pool was observed in either image. In Fig. 6, the results indicate that the computational model's predictions are deviated from the experimental outcomes by merely 8.49% and 4.8%, respectively, for the width and depth of the melt pool with circular speed of 55 mm/s for laser powers of 2700 W and 2750 W, respectively. Possible explanations for the difference between the simulation and experimental data in Figs. 5 and 6 may include complex thermal gradients, fluid flow patterns, and surface interactions that are challenging to accurately simulate in characterizing the recoil pressure, surface traction, diffusion, etc. in the modelling. To better understand and validate the observed difference, future investigations will be conducted to directly observe the real-time phenomena.

Thus, the basic validity of the CFD simulation model is confirmed. These findings highlight the reliability and efficacy

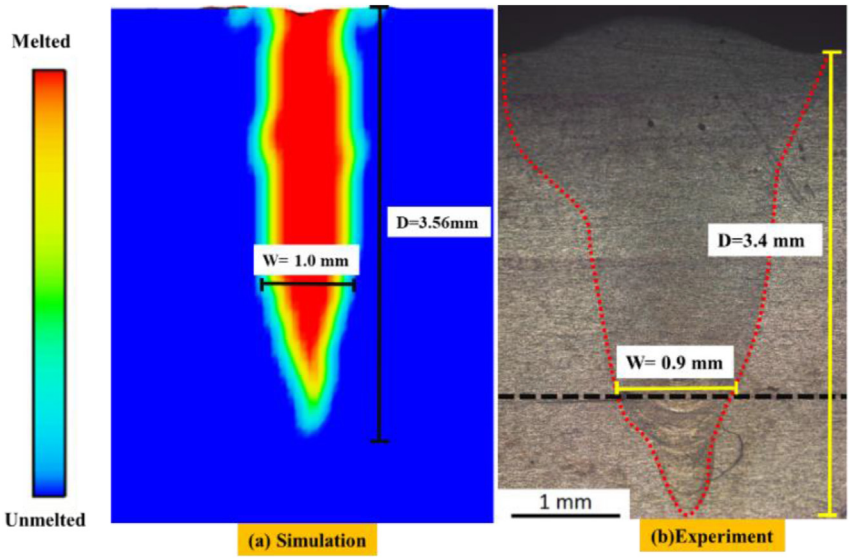


Fig. 5 – Comparison of melt pool profiles from simulation and experiment. (a) Simulation image for the cross-section of the melt pool, and (b) Optical Microscopy image of the cross-section of the melt pool.

of the computational model for predicting the geometry of the melt pool in Al–Cu lap joints under varied laser processing parameters. Overall, the comparative analysis provides compelling evidence for the high degree of accuracy of the computational model and its utility as a valuable tool for optimizing laser processing parameters and enhancing the quality of Al–Cu lap joints.

4. Study of optimal design in parameters

4.1. *Surrogate modeling of laser welding process*

Tran and Lo [23] have devised a novel approach to minimize the computational burden of investigating the impact of laser power and scanning speed on melt pool generation in the selective laser melting (SLM) process. Instead of conducting hundreds of finite element (FE) simulations, they employed a circle packing design method to select 48 representative combinations of scanning speed and laser power. These simulations generated data that was then used to train artificial neural network (ANN) models capable of predicting the melt pool geometry and peak temperature for any combination of scanning speed and laser power within the feasible design space. This approach offers a solution to the time-intensive process of performing hundreds of simulations, as the ANN models can be used to quickly and accurately predict the melt pool characteristics based on the laser processing parameters.

The study aimed to investigate the effects of laser power and circular speed on the melt pool geometry and Cu

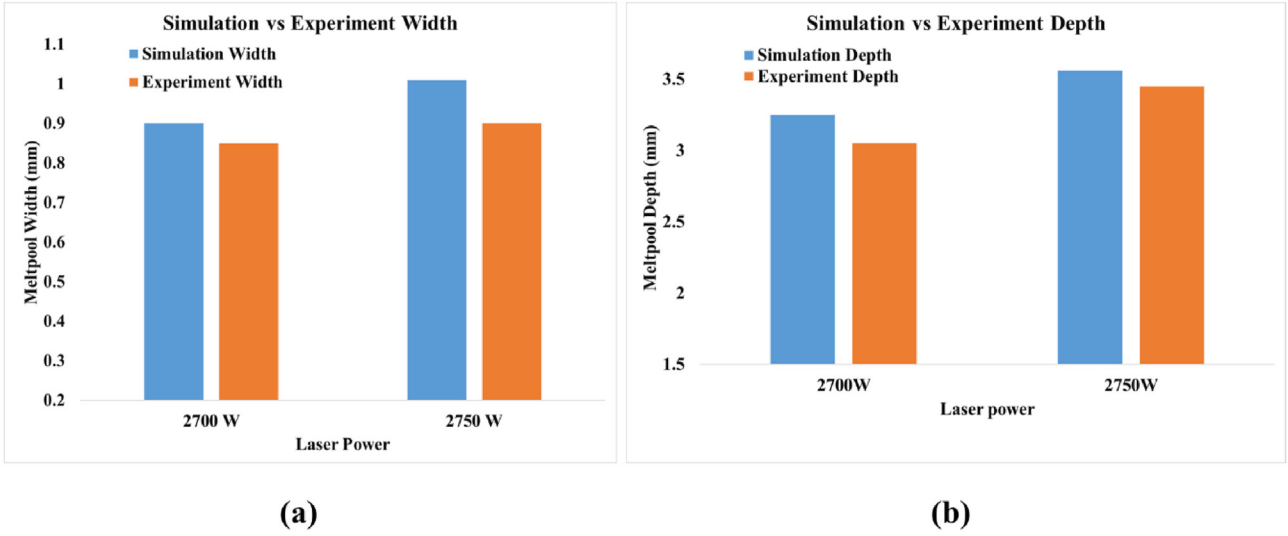


Fig. 6 – Comparison of results for simulation and experiment: (a) Melt pool width and (b) Melt pool depth with circular speed of 55 mm/s for laser powers of 2700 W and 2750 W, respectively.

concentration in an Al–Cu lap joint using laser welding. To achieve this, a circle packing design approach was employed to determine a set of sampling points in the design space, which included the laser power and tangential speed used in the laser welding process. The design space was defined as the range of laser power from 2700 to 3300W and circular speed from 45 to 75 mm/s (see Fig. 7). After the circle packing design algorithm was executed, a total of 39 sample points were obtained. These sample points were then used as input for a CFD simulation model to calculate the corresponding values for the melt pool geometry and Cu concentration. The simulation results were then used to train ANN models to predict the melt pool geometry and Cu concentration for any combination of laser power and circular speed within the design space. To build processing maps for the melt pool geometry and Cu concentration, the trained ANN models were used to predict the corresponding results for a large number of parameter sets, chosen from the feasible design space with increments of 0.5 mm/s and 10W in the circular speed and laser power axis, respectively (see Fig. 8). However, each simulation run takes on average around 16 h to complete when using an AMD Ryzen Thread ripper 3970X 32-Core processor (CPU), and hence such an approach is clearly infeasible.

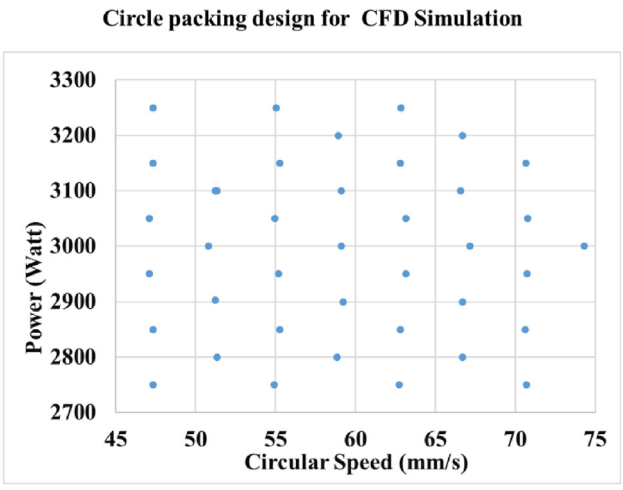


Fig. 7 – Circle packing design for Simulation.

4.2. *Optimization of CW laser welding parameters for high Al–Cu joint strength with different criteria*

In this study, the optimization of processing parameters for the laser welding process was achieved by using ANN models that were trained on the simulation results of the laser welding process. The trained ANN models were then used to screen through various processing maps to identify the parameters that result in a high mechanical strength. To validate the feasibility of these parameters, a series of experimental trials were conducted, and the results were analyzed based on three quality criteria - melt pool interface width, melt pool penetration depth, and Copper concentration. This approach helped to identify the most suitable processing parameters that could produce a high-quality laser-welded joint. Therefore, three quality criteria are designed in order to get the optimal parameters.

4.2.1. *Melt pool interface width criterion*

The width of the melt pool interface is a critical factor in determining the quality of the Al–Cu joint. As per the findings in Ref. [12], a larger width of the melt pool interface is desirable to ensure a good electrical contact when welding Al–Cu joints with lower peak power. In the present study, a melt pool interface width criterion of 0.4 mm–0.95 mm was established as the green zone for acceptable joint quality. The trained ANN was used to predict the melt pool interface width for 3722 samples in the feasible design space, as shown in Fig. 9(a). The processing map was then screened using the established criterion value to identify the optimal parameters. The outcome of the screening is shown in Fig. 9(b), with parameters resulting in an acceptable interface width being marked in green. The blue zone of the filtered processing map represents parameters resulting in an interface width less than 0.4 mm, while the red zone represents parameters resulting in an interface width greater than 0.95 mm.

4.2.2. *Melt pool penetration depth criterion*

Jarwitz et al. [5] have demonstrated that a wider melt pool interface width and lower penetration depth can effectively reduce the electrical contact resistance of dissimilar Al–Cu welds. This reduction in electrical resistance has been found to be correlated with increased mechanical strength in numerous studies [8,24]. Therefore, the combination of a wider melt pool interface width and lower penetration depth can potentially lead to stronger Al–Cu welds with lower electrical contact resistance is beneficial in reducing the electrical contact resistance of dissimilar Al–Cu welds. In the study by Dimatteo et al. [25], it was observed that the mechanical strength of Al–Cu welds decreases with increasing depth of penetration. To evaluate the quality of the welds, a criterion was established based on the depth of penetration. The criterion classified penetration depths less than 2.50 mm as blue, depths between 2.50 mm and 4 mm as green, and depths greater than 4 mm as red. The original processing map for melt pool penetration depth was obtained using an artificial neural network (ANN) and is presented in Fig. 10(a). To filter the map based on the established criterion, a color-coded map was created. The red region in the filtered map represents excessive penetration depth, the green region represents an acceptable depth, and the blue region indicates an excessively narrow melt pool depth as shown in Fig. 10(b). It is important to note that the goal of the study was to identify regions with acceptable penetration depth, which corresponds to the green region in the filtered map. The blue and red regions indicate areas of the weld that may have reduced mechanical strength due to excessive or inadequate penetration depth, respectively. Overall, the study provides a useful approach for evaluating the quality of Al–Cu welds based on melt pool penetration depth.

4.2.3. *Cu concentration criterion*

The control of brittle IMC formation during dissimilar welding of aluminum and copper presents a significant challenge. According to Huang et al. [17], a concentration of copper greater than 50% by weight in the aluminum region of the melt

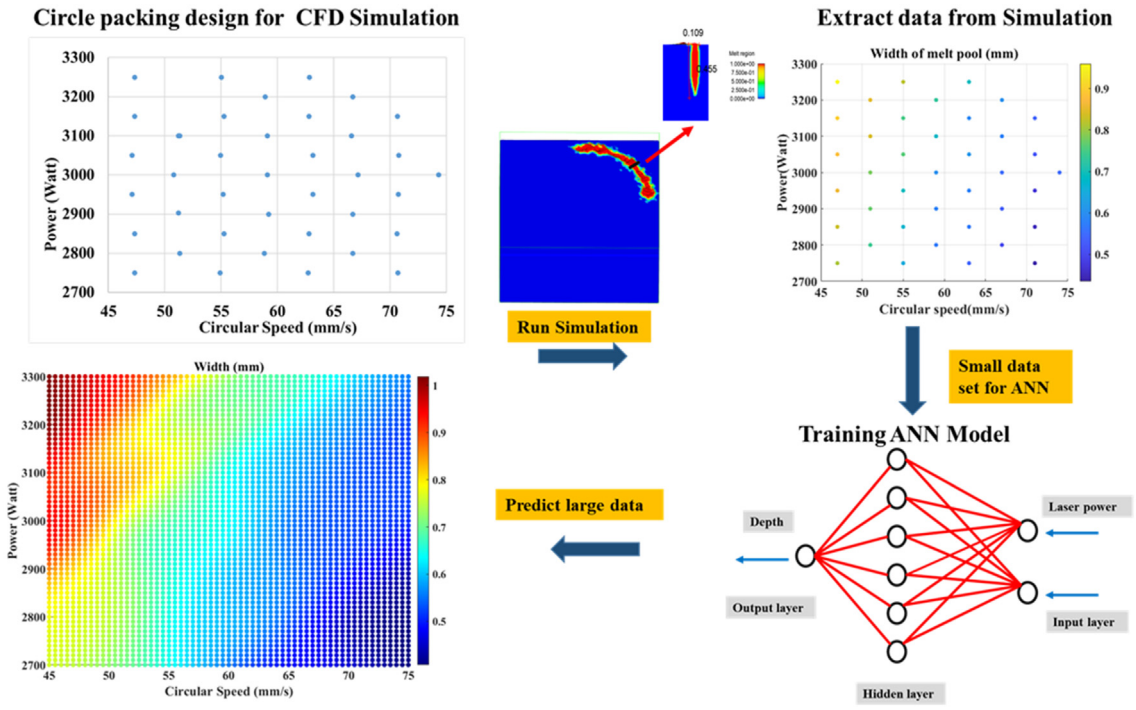


Fig. 8 – Flow chart representation for main steps involved in surrogate model.

pool leads to the fusion zone's susceptibility to cracking during the solidification phase. Furthermore, in accordance with the Al–Cu phase diagram, the formation of hard and brittle IMCs such as Al_2Cu and AlCu occurs at copper concentrations exceeding 50 wt% [8,26,27]. These IMCs, in turn, promote the initiation and propagation of cracks in the fusion zone. Hence, the prevention of IMC formation is crucial for achieving successful dissimilar welding of aluminum and copper.

In the present investigation, the simulation results for the distribution of Cu in the weld pool of the Al–Cu lap joint sample processed using a laser power of 3200W and a circular speed of 50 mm/s are presented in Fig. 11. It can be observed from the figure that the upper part of the weld pool is predominantly composed of Al, however, there is a small presence of Cu due to the fluid flow from the bottom Cu plate. On the other hand, a small amount of Al is present in the lower

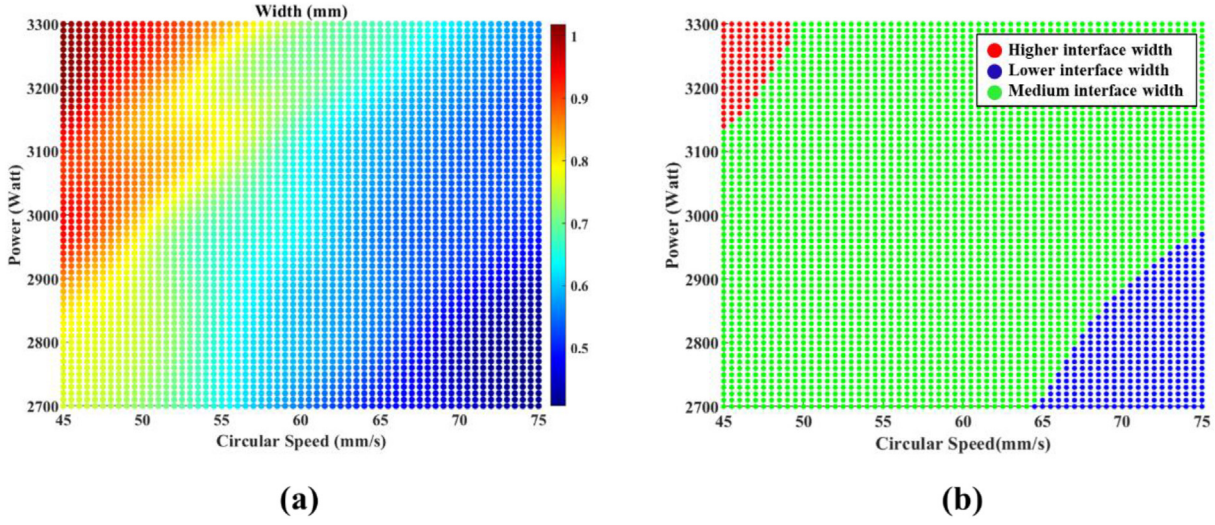


Fig. 9 – (a) Processing map for meltpool width produced by ANN, and (b) processing map after filtration. Red region: higher interface width, green region: medium interface width; and blue region: lower interface width. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

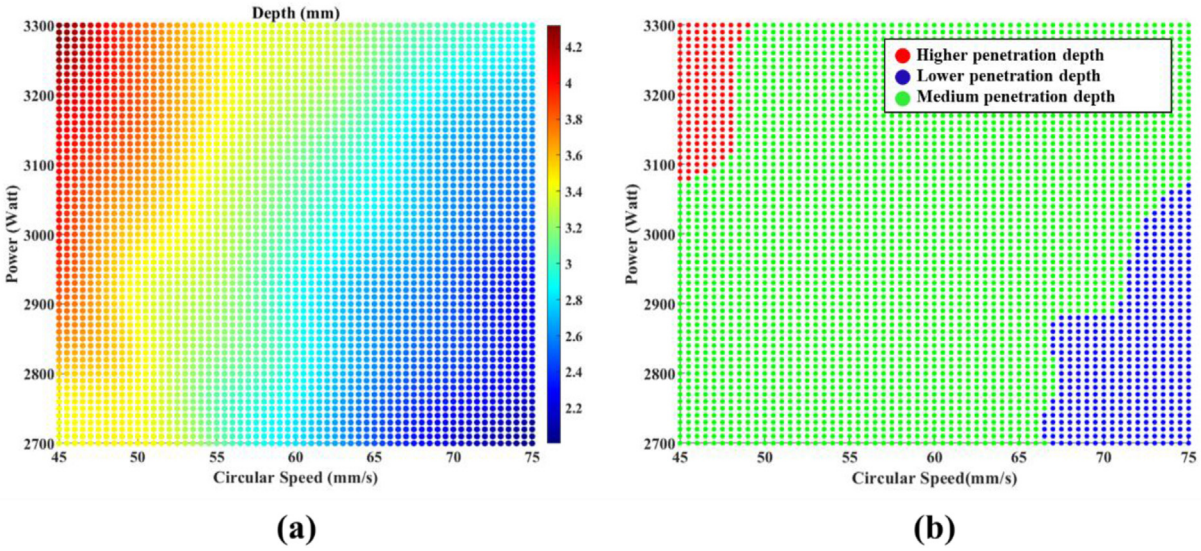


Fig. 10 – (a) Processing map for melt pool depth produced by ANN, and (b) processing map after filtration. Red region: higher penetration depth; green region: medium penetration depth; and blue region: lower penetration depth. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

part of the weld pool, which originated from the upper Al plate. To determine the highest concentration of Cu within the Al region of the weld pool, a processing map was created using an ANN and is displayed in Fig. 12(a). The purpose of the processing map is to provide an overview of the relationships between the laser processing parameters and the resulting Cu concentration within the Al region of the weld pool. The results obtained after screening the processing map using a criterion value of 50 wt% for the Cu concentration are shown in Fig. 12(b). The processing parameter settings that resulted in an acceptable level of Cu concentration are depicted in green (as implied in Refs. [8,26,27]).

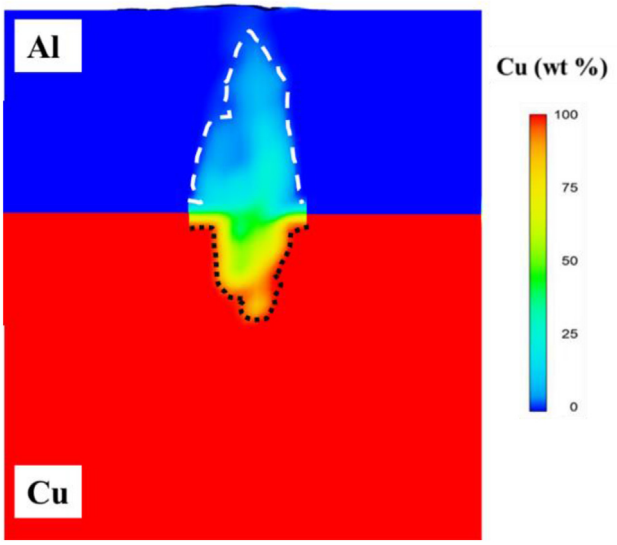


Fig. 11 – Simulation cross section of Cu concentration for Al–Cu joint.

4.3. *Optimization of the processing map region*

Fig. 13 depicts the processing map that was created by superimposing the screened maps in Fig. 9(b), 10(b) and 12(b), respectively. The process parameters are a critical factor in determining the quality of the Al–Cu lap joint produced through dissimilar circular laser welding. The green region in the graph depicts the optimal processing parameters that lead to the production of high-quality Al–Cu joints, characterized by enhanced mechanical strength, and minimal formation of IMC. On the other hand, the red region in the graph represents the process conditions that result in excessive penetration depth, thereby increasing the risk of cracking and porosity formation in the fusion zone. This excessive penetration depth is a result of the high laser power and circular laser speed used during the welding process. The blue region in the graph represents the process parameters that result in a limited melt pool interface width and depth. This narrow interface width can lead to a reduced bond strength between the Al and Cu materials. Furthermore, the high concentration of copper in the fusion zone, which is a result of the welding parameters in the red region, can lead to the formation of pores and cracks. This can further weaken the overall mechanical properties of the joint and reduce its performance in practical applications. Therefore, it is crucial to carefully choose the process parameters to achieve the desired quality of the Al–Cu lap joint and to minimize the risk of defects and poor performance.

To validate the screened processing map, an experimental verification was conducted to confirm the effectiveness of the selected parameters from the green region. The validation process involved selecting four points randomly from the optimal region of the map and conducting welding trials using those parameters. Additionally, a small number of points

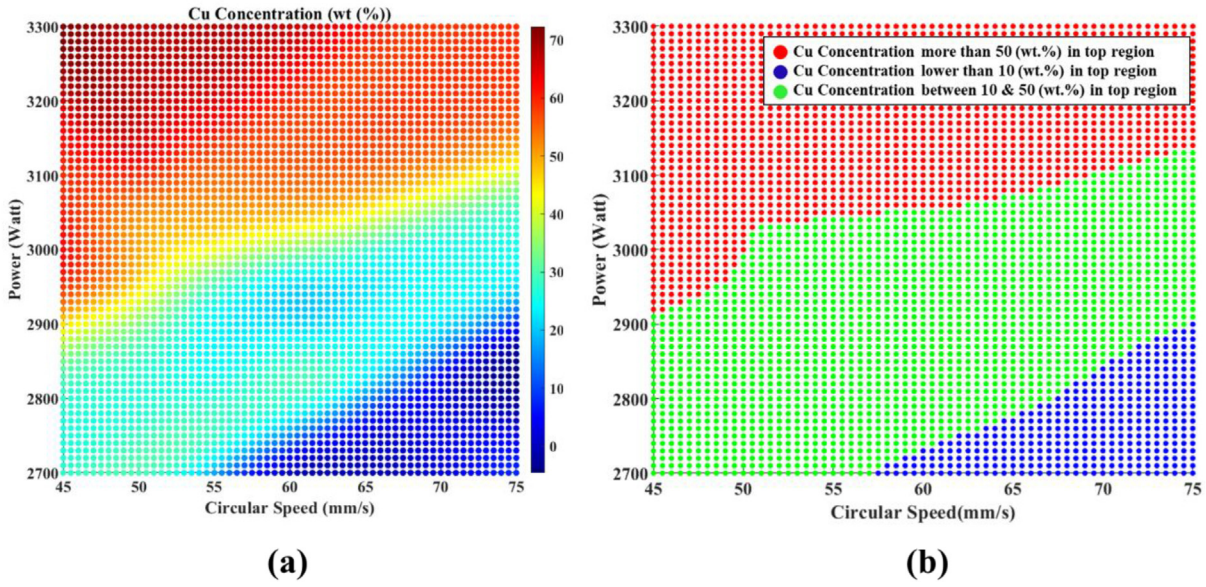


Fig. 12 – (a) Processing map for Cu-concentration produced by ANN, and (b) filtered processing map screened by Cu concentration criterion. Blue region: Cu concentration lower than 10 wt%; red region: Cu concentration greater than 50 wt% and green region: Cu concentration greater than 10 wt% and lower than 50 wt%. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

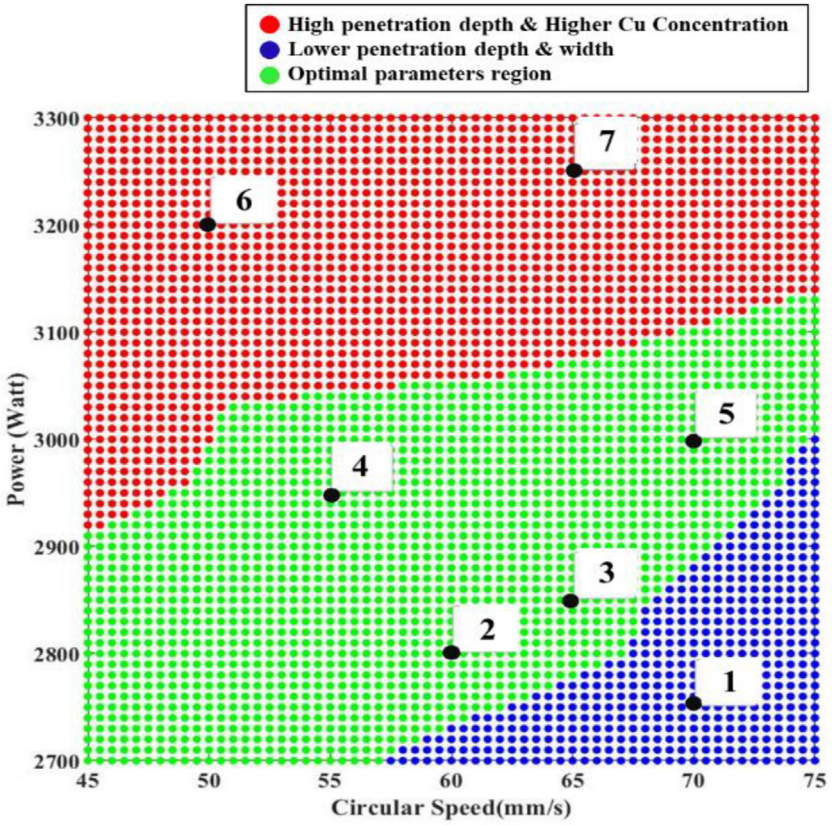


Fig. 13 – Optimized processing map after filtration with three quality criterion i.e. width of melt pool, depth of met pool and Cu-concentration.

Table 2 – Experimental parameters chosen for validation trials.

Sample No.	Power (W)	Circular speed (mm/s)
1	2750	70
2	2800	60
3	2850	65
4	2950	55
5	3000	70
6	3200	50
7	3250	65

were chosen from other regions of the map for comparison purposes as shown in Table 2. The experimental trials were performed to assess the quality of the produced joints and to validate the accuracy of the screened processing map. The results of the experimental validation were used to confirm the feasibility of the selected parameters, ensuring that they result in high-quality Al–Cu lap joints.

5. Experimental results

5.1. Top surface analysis

Fig. 14 presents images of the specimen's top surface after welding, produced using different processing conditions selected from various regions of the filtered processing map as shown in Fig. 13. These images clearly show that samples processed within the green zone of the optimized parameters exhibit a smooth surface, which meets the required specifications from the manufacturer. On the other hand, parameters from the blue region of the processing map also exhibit a smooth top surface, as the laser energy is not sufficient to completely melt the aluminum sheet. However, samples processed in the red zone exhibit a pores on the top surface due to the excessive laser energy used, which leads to the formation of cracks [28] at the interface and decreases the joint strength. As a result, the optimized parameters in this study have been shown to provide reliable results and ensure smooth surfaces, which are critical in maintaining the efficiency and performance of laser welding.

5.2. Electron dispersive spectroscopy (EDS) analysis results

Laser welding processes generate a significant thermal gradient that facilitates the development of diverse microstructures and intermetallic (IMC) phases [29]. Two lap-joint samples were selected for Energy-Dispersive X-ray Spectroscopy (EDS) analysis. These samples were processed under conditions chosen from the green and red regions of the filtered processing map. The SEM/EDS analysis was performed on these samples. In Fig. 15, it shows the SEM/EDS analysis results of sample Number 4 under optimal

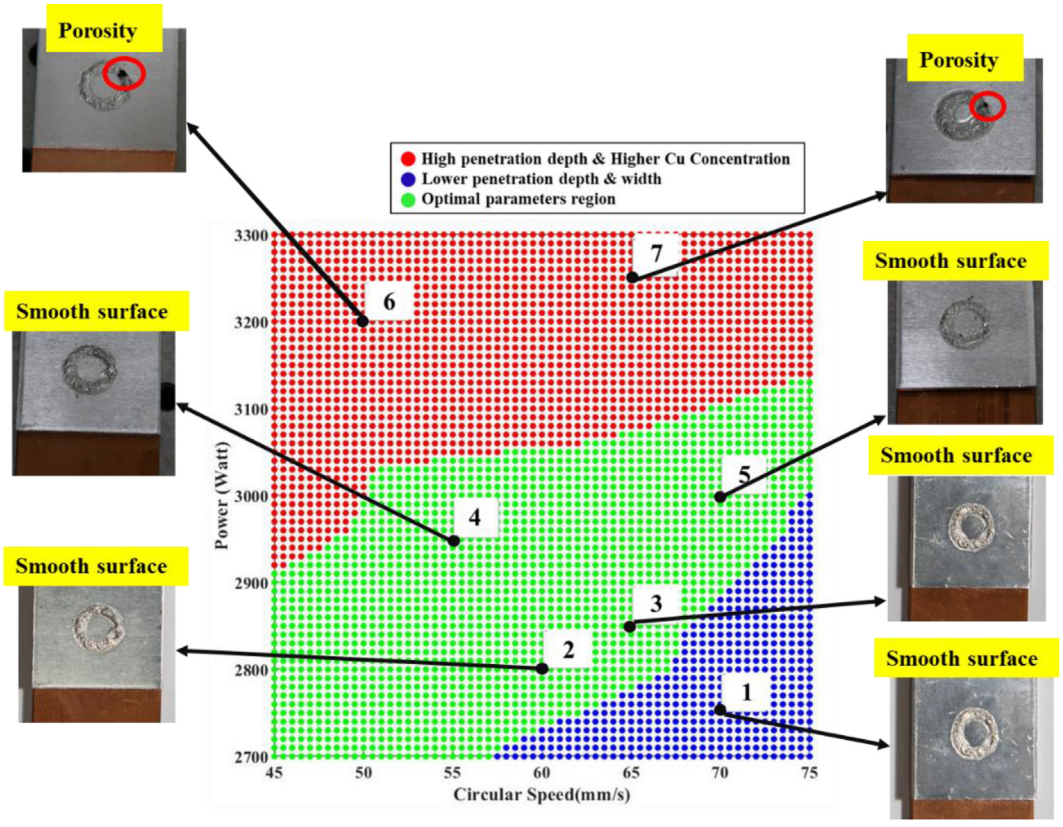
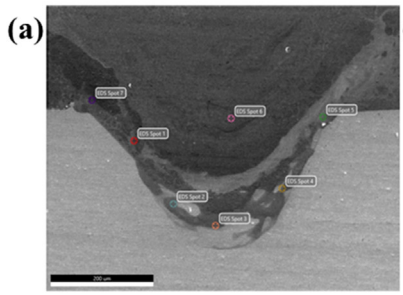


Fig. 14 – Top surface images of welded joint produced using different processing conditions selected from various regions of filtered processing map.



(b)

Elements	Al (at%)	Cu (at%)	Possible Phases (Ref.)
EDS spot 1	87	13	Al Solid Solution [29,30]
EDS spot 2	58	42	AlCu [31]
EDS spot 3	65	35	Al_2Cu [31,32]
EDS spot 4	60	40	AlCu [31]
EDS spot 5	67	33	Al_2Cu [31,32]
EDS spot 6	87	13	Al Solid Solution [29,30]
EDS Spot 7	85	15	Al Solid Solution [29,30]

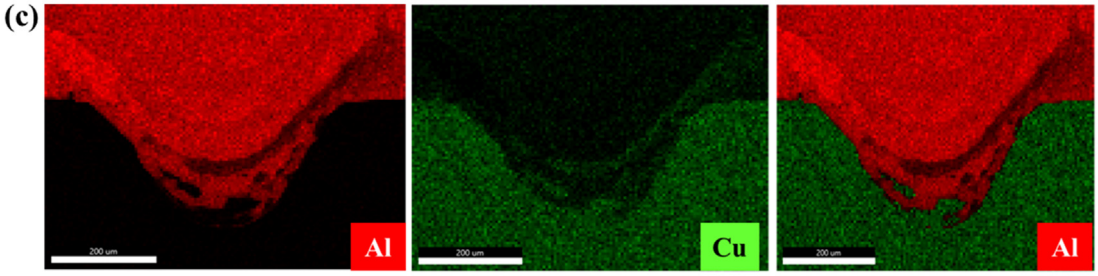
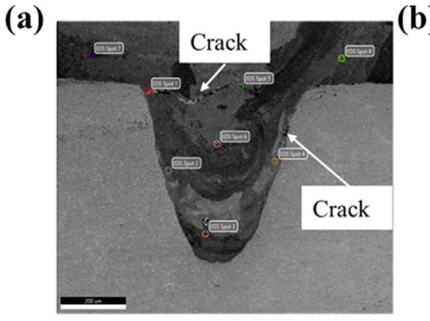


Fig. 15 – (a) Results of EDS scanning for met pool cross section of sample produced with optimal parameters, (b) EDS phase compositions at selected scanning spots, and (c) Al and Cu element distributions in weld pool.

processing condition marked as green in Fig. 13 with the laser power of 2950W and circular speed of 55 mm/s. Observing the melt pool's boundary region reveals that Spots 1, 6, and 7 may contain Al solid solution [29,30]. Spots 2 and 4 appear to be of AlCu [31] phase, indicating a uniform distribution of Al and Cu elements at the weld interface. Spots 3 and 5, which may consist of Al_2Cu phase, have a lower Cu concentration than Al_4Cu_9 phase and thus are less brittle [31,32]. The fusion zone remains crack-free, as demonstrated in Fig. 15. According to the EDS mapping results depicted in Fig. 15(c), Al and Cu elements are evenly distributed throughout the melt pool, suggesting enhanced mechanical strength of the joint.

As presented in Fig. 16, it shows the SEM/EDS analysis results of sample Number 7 under non-optimized processing condition marked as red in Fig. 13 with the laser power of 3250W and circular speed of 65 mm/s. Spots 1 and 4 possibly consist of Al_3Cu_4 which is highly brittle and prone to cracking [33]. As can be seen in Fig. 16, some cracks are visible possible and the reason possible might be the excessive Cu concentration. Spots 2 and 3 consists of Al_2Cu which has an adverse effect on the strength of the joint [29,30]. The presence of a



(b)

Elements	Al (at%)	Cu (at%)	Possible Phases (Ref.)
EDS spot 1	20	80	Al_4Cu_9 [33]
EDS spot 2	65	35	Al_2Cu [29,30]
EDS spot 3	70	30	Al_2Cu [29,30]
EDS spot 4	30	70	Al_4Cu_9 [33]
EDS spot 5	45	55	AlCu [31]
EDS spot 6	75	25	Al_3Cu [31,32]
EDS Spot 7	70	30	Al_3Cu [31,32]
EDS Spot 8	74	26	Al_3Cu [31,32]

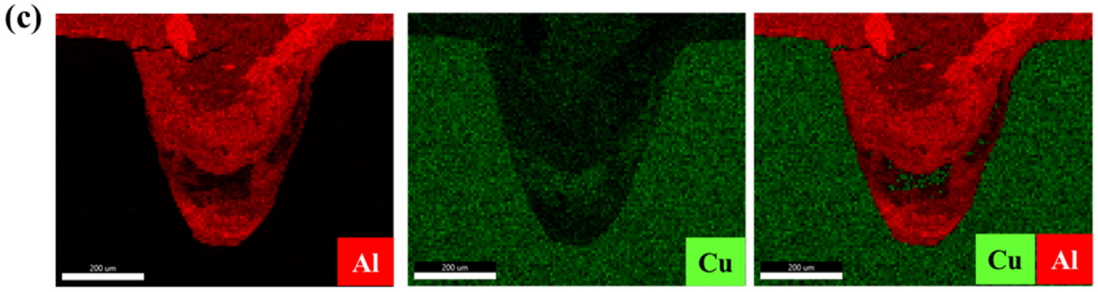


Fig. 16 – (a) Results of EDS scanning for met pool cross section of sample produced with non-optimal parameters, (b) EDS phase compositions at selected scanning spots, and (c) Al and Cu element distributions in weld pool.

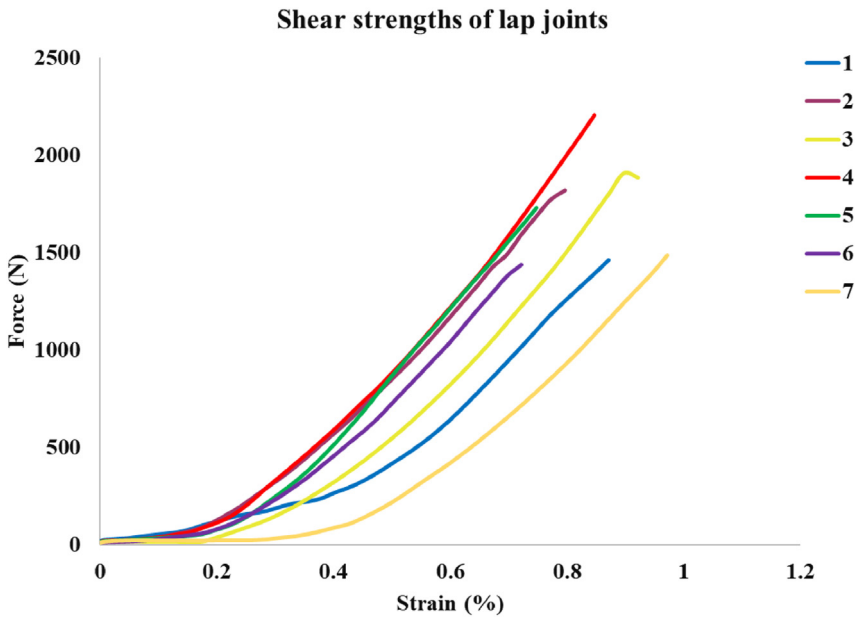


Fig. 17 – Shear strength of lap joints produced using different processing conditions.

substantial quantity of intermetallic compound (IMC) phase within the molten pool elevates the probability of solidification cracking [29,30]. The formation of visible pores and cracks within the fusion zone is a predictable outcome, as depicted in Fig. 13. Specifically, the red region exhibits a substantial amount of cracking and porosity. Spots 6, 7, and 8 possibly consists of Al_3Cu [31,32]. The non-uniform distribution of the Al and Cu contents in the melt pool is revealed by the EDS mapping results in Fig. 16(c), which leads to a weaker joint strength.

Kermanidis et al. [34] examined the use of a long pulse laser for spot welding copper tubes to aluminum foils. The study presents evidence of a direct relationship between the formation of IMCs and the heat input applied during welding. The results indicated that an increase in heat input leads to a corresponding rise in IMC formation. Authors in Refs. [12,13] have utilized pulse laser technology to manage heat input by an optimal process for successfully reducing the occurrence of brittle IMCs in the fusion zone. Similarly, as in Ref. [13], the current study shows the effectiveness of parameter optimization in controlling heat input during CLW with a circular strategy, thereby minimizing the formation of IMCs. It is found that IMCs issues occurring in Ref. [13] using PLW and the current study using CLW are quite similar with few difference. It is concluded that the parameter optimization effectively plays an important key role in welding dissimilar materials.

Table 3 – Mechanical properties of Al–Cu test pieces.

Sample No.	Yield Strength (N)	Ultimate Tensile Strength (N)
1	1443	1453
2	1810	1810
3	1912	1912
4	2208	2208
5	1730	1730
6	1355	1444
7	1481	1481

5.3. *Tensile test for mechanical properties*

The results of the experiment on maximum shear strengths, as depicted in Fig. 17, indicate that the lap joints produced using the optimal processing parameters selected by the proposed method exhibit the highest shear strength. This conclusion was supported by the measurement results for the yield strength (YS) and ultimate tensile strength (UTS) of the lap joints, presented in Table 3. The data shows that the weld strength of the lap joints increases from 1443 N for the Sample in the blue region (2750W and 70 mm/s) to 2208 N in the green region (2950W and 55 mm/s), before decreasing to 1355 N and 1481 N in the red region (3200W and 50 mm/s; 3250W and 65 mm/s). These results demonstrate that the proposed method effectively selects the optimal processing parameters to produce the highest shear strength among the lap joints tested. The findings presented in Table 3 reveal a notable enhancement in the shear force values of aluminum and copper interconnects, achieved through the optimization of parameters in this study. The results of shear force for aluminum and copper joints in current study surpass those reported in prior studies [12,13,15,24,35] from the global perspective, particularly in terms of dissimilar laser welding, of course it is also related to the bonding area of joints. These results highlight the importance and relevance of the current study as it contributes to improving the shear strength of dissimilar materials by suppressing the highly brittle intermetallic compounds formation at the fusion zone. Such improvements are crucial in guiding the development of future interconnect designs and specifications.

6. Conclusions

This study presents a new approach for determining the best processing parameters for making dissimilar metal joints between aluminum and copper using continuous laser welding with a circular strategy. The method involves using CFD simulations to model the welding process and predict the resulting joint quality. A surrogate modeling approach is then used to identify the optimal values for parameters such as laser power and welding speed that result in the highest quality joint.

The study has made the following major contributions and findings.

- The CFD-based numerical model proposed in this study demonstrates a high degree of accuracy in predicting the shape of the melt-pool across a range of laser power and circular welding speed within the feasible design space.
- Artificial neural network models were trained using the validated finite element model, and these models can predict the melt pool interface width, penetration depth, and copper concentration for any combination of processing conditions. The optimal processing conditions were determined using quality criteria based on melt pool width, depth, and copper concentration.
- Experimental trials confirmed the validity of the proposed optimal processing conditions, and it was found that the Al–Cu joint showed excellent mechanical properties under these conditions with a maximum shear strength of 2200 N. The results of shear force for aluminum and copper joints in this study demonstrate better performance compared to previous studies.
- Microstructural analysis and tensile-shear strength testing showed that maximum shear strength was achieved when optimal processing conditions were used, resulting in a low degree of intermixing and the absence of large intermetallic structures in the weld seam. The effectiveness of parameter optimization in controlling heat input during CLW with a circular strategy, thereby minimizing the formation of IMCs, was also demonstrated.

This study offers a valuable contribution to the welding and materials science communities, as it presents a novel approach to address a common challenge in the welding of dissimilar metals. Specifically, the study demonstrates the potential of continuous laser welding in producing battery tab connections. However, despite the promising results, the existence of metallurgical defects, such as poor bonding, formation of brittle intermetallic phases, and cracking, still poses a significant obstacle.

To further advance the field, future studies should focus on conducting a more detailed analysis of the stability of these connections over time, particularly under cyclic loads resulting from temperature changes or electrical current flow. This would provide a better understanding of how the connections would perform under repeated stress and how they might degrade over time. By doing so, researchers would be better equipped to assess the durability and reliability of the connections, which are critical factors in the success of any welding application.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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